

7.5 Sampling distributions of the estimates

A point estimate is of little use without a measure of its uncertainty. Under the second-moment assumptions we know the mean and variance of $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ and the mean of s^2 , but not their distributions. Under normality we know everything. The estimates of the estimable functions are jointly normal, the residual sum of squares is a scaled chi-squared variable, and the two are independent. These three facts are what the whole of Part III rests on. Tests, confidence intervals, confidence ellipsoids and prediction intervals are all built from them. The machinery was assembled in [Chapter 3](#) and [Chapter 4](#). Here we apply it.

7.5.1 The main theorem

Theorem 7.5.1. Sampling distributions in the normal linear model

Assume Eq. 7.3.1 with $r = \text{rank}(\mathbf{X}) < n$. Let $\boldsymbol{\Lambda}$ be $p \times q$ with $\mathcal{C}(\boldsymbol{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$, let $(\mathbf{X}^\top \mathbf{X})^-$ be any generalized inverse, and let $s^2 = \text{SSE}/(n - r)$.

- (a) $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} \sim N_q(\boldsymbol{\Lambda}^\top \boldsymbol{\beta}, \sigma^2 \boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\Lambda})$. In particular $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{M})$, and if $\text{rank}(\mathbf{X}) = p$, $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$.
- (b) $\text{SSE}/\sigma^2 = (n - r)s^2/\sigma^2 \sim \chi^2(n - r)$.
- (c) $\mathbf{M}\mathbf{Y}$ and SSE are independent. Hence $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}$, and every function of the fitted values, is independent of s^2 .
- (d) $E(s^2) = \sigma^2$ and $\text{Var}(s^2) = 2\sigma^4/(n - r)$.

Proof. Write $\boldsymbol{\Lambda} = \mathbf{X}^\top \mathbf{P}$ for an $n \times q$ matrix $\mathbf{P}$. As in the proof of Theorem 7.2.1(d), $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{P}^\top \mathbf{M}\mathbf{Y}$ for every least squares estimate $\hat{\boldsymbol{\beta}}$.

- (a) By Theorem 3.3.1, $\mathbf{P}^\top \mathbf{M}\mathbf{Y}$ is normal with mean $\mathbf{P}^\top \mathbf{M}\mathbf{X}\boldsymbol{\beta} = \mathbf{P}^\top \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\Lambda}^\top \boldsymbol{\beta}$ and covariance $\sigma^2 \mathbf{P}^\top \mathbf{M}\mathbf{M}^\top \mathbf{P} = \sigma^2 \mathbf{P}^\top \mathbf{M}\mathbf{P} = \sigma^2 \boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\Lambda}$, by Theorem 6.3.6. The special cases are $\boldsymbol{\Lambda} = \mathbf{X}^\top$, with $\mathbf{P} = \mathbf{I}$, and $\boldsymbol{\Lambda} = \mathbf{I}_p$ in full rank.
- (b) $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y}$, and $\mathbf{I} - \mathbf{M}$ is symmetric and idempotent of rank $n - r$ (Theorem 6.3.3 and Proposition 6.3.4). By Theorem 4.3.2(a), $\text{SSE}/\sigma^2 \sim \chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2)$, and the noncentrality is zero because $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$.
- (c) Apply Theorem 4.4.1 with $\mathbf{B} = \mathbf{M}$, $\mathbf{A} = \mathbf{I} - \mathbf{M}$ and $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$. Then $\mathbf{B}\boldsymbol{\Sigma}\mathbf{A} = \sigma^2 \mathbf{M}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$, so $\mathbf{M}\mathbf{Y}$ and $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y}$ are independent. Functions of independent random vectors are independent.
- (d) A $\chi^2(k)$ variable has mean k and variance $2k$ (Theorem 4.1.2(b) with $\gamma = 0$). So $E \text{SSE} = \sigma^2(n - r)$ and $\text{Var}(\text{SSE}) = 2\sigma^4(n - r)$. Divide by $n - r$ and $(n - r)^2$.

■

The proof is short because the work was done earlier. It is worth seeing where each ingredient comes from geometrically. The fitted vector $\mathbf{M}\mathbf{Y}$ and the residual vector $(\mathbf{I} - \mathbf{M})\mathbf{Y}$ are projections of $\mathbf{Y}$ onto orthogonal subspaces. For a spherical normal vector, orthogonal projections are independent (Theorem 4.4.4). Everything about $\boldsymbol{\beta}$ lives in the first projection. The second is pure noise, and its squared length, measured in units of σ^2 , counts $n - r$ independent squared standard normals.

The independence in (c) has no analogue under the second-moment assumptions. If the errors are independent with zero third moment, $\lambda^\top \hat{\beta}$ and s^2 are still uncorrelated, but in general they are not independent. The simulation at the end of the section shows the difference.

7.5.2 Standardizing: the t statistic

The distribution in (a) involves the unknown σ . Since $E(s^2) = \sigma^2$, the matrix $s^2 \Lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \Lambda$ is an unbiased estimate of $\text{Cov}(\Lambda^\top \hat{\beta})$, and in full rank $s^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ is unbiased for $\text{Cov}(\hat{\beta})$; this needs only the second-moment assumptions. The square roots of its diagonal entries are the standard errors reported by regression software. Replacing σ by s changes the normal distribution into a t distribution, and independence is what makes the change exact.

Corollary 7.5.2. The t statistic for an estimable function

Assume Eq. 7.3.1 with $r < n$, let $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ with $\lambda \neq \mathbf{0}$, and put $\text{se}(\lambda^\top \hat{\beta}) = s \sqrt{\lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda}$. For every constant d ,

$$T_d = \frac{\lambda^\top \hat{\beta} - d}{\text{se}(\lambda^\top \hat{\beta})} \sim t(n - r, \delta), \quad \delta = \frac{\lambda^\top \beta - d}{\sigma \sqrt{\lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda}}.$$

In particular $T = (\lambda^\top \hat{\beta} - \lambda^\top \beta) / \text{se}(\lambda^\top \hat{\beta}) \sim t(n - r)$, and

$$\Pr\{|\lambda^\top \hat{\beta} - \lambda^\top \beta| \leq t_{n-r, \alpha/2} \text{se}(\lambda^\top \hat{\beta})\} = 1 - \alpha, \quad (7.5.1)$$

where $t_{n-r, \alpha/2}$ is the upper $\alpha/2$ point of $t(n - r)$.

Proof. Put $c^2 = \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda$. Then $c > 0$, since $c^2 = \|\mathbf{M}\mathbf{p}\|^2$ with $\mathbf{X}^\top \mathbf{M}\mathbf{p} = \lambda \neq \mathbf{0}$. By Theorem 7.5.1(a), $Z = (\lambda^\top \hat{\beta} - d) / (\sigma c) \sim N(\delta, 1)$. By (b), $V = (n - r)s^2 / \sigma^2 \sim \chi^2(n - r)$, and by (c) Z and V are independent. Now $T_d = Z / \sqrt{V / (n - r)}$, which is $t(n - r, \delta)$ by Definition 4.1.3. With $d = \lambda^\top \beta$, $\delta = 0$, and Eq. 7.5.1 restates $\Pr\{|T| \leq t_{n-r, \alpha/2}\} = 1 - \alpha$. ■

Eq. 7.5.1 is the familiar confidence interval $\lambda^\top \hat{\beta} \pm t_{n-r, \alpha/2} \text{se}$, and T_d is the statistic for testing $\lambda^\top \beta = d$. Its noncentrality δ governs the power. Chapter 11 and Chapter 12 develop both. For the air-flow coefficient in the stack loss regression this interval was (0.431, 1.000) (Example (Profile likelihood for the air-flow coefficient)). Two features of the corollary are easy to overlook. The degrees of freedom are $n - r$, the dimension of the residual space. They are not n , and when $\mathbf{X}$ is rank deficient they are not $n - p$ either. And the interval is exact for every sample size, which no large-sample argument delivers.

7.5.3 Several functions at once

For a vector of estimable functions, the natural summary of Theorem 7.5.1(a) is a quadratic form. The covariance matrix $\Lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \Lambda$ may be singular when the columns of Λ are linearly dependent, so a generalized inverse is needed.

Corollary 7.5.3. A chi-squared pivot for several functions

In the setting of [Theorem 7.5.1](#), let $\mathbf{W} = \mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{\Lambda}$, let $\mathbf{W}^-$ be any generalized inverse of $\mathbf{W}$, and put $\mathbf{D} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{\Lambda}^\top \boldsymbol{\beta}$. Then

$$Q = \frac{\mathbf{D}^\top \mathbf{W}^- \mathbf{D}}{\sigma^2} \sim \chi^2(\text{rank}(\mathbf{\Lambda})),$$

Q does not depend on the choice of $\mathbf{W}^-$ (with probability one), and Q is independent of SSE. Consequently

$$\frac{\mathbf{D}^\top \mathbf{W}^- \mathbf{D} / \text{rank}(\mathbf{\Lambda})}{s^2} \sim F(\text{rank}(\mathbf{\Lambda}), n - r).$$

Proof. By [Theorem 7.5.1\(a\)](#), $\mathbf{D} \sim N_q(\mathbf{0}, \sigma^2 \mathbf{W})$. The quadratic form is unchanged if $\mathbf{W}^-$ is replaced by its symmetric part $\frac{1}{2}(\mathbf{W}^- + \mathbf{W}^{-\top})$, which is again a generalized inverse of the symmetric matrix $\mathbf{W}$ (Proposition 1.3.3(d)). So we may assume $\mathbf{W}^-$ symmetric. Apply Theorem 4.3.2(c) with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{W}$, $\boldsymbol{\mu} = \mathbf{0}$ and $\mathbf{A} = \mathbf{W}^- / \sigma^2$. Conditions (ii) and (iii) hold trivially because $\boldsymbol{\mu} = \mathbf{0}$. Condition (i) is $\boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \sigma^2 \mathbf{W} \mathbf{W}^- \mathbf{W} \mathbf{W}^- \mathbf{W} = \sigma^2 \mathbf{W} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}$, by two uses of $\mathbf{W} \mathbf{W}^- \mathbf{W} = \mathbf{W}$. So $Q \sim \chi^2(\text{tr}(\mathbf{A} \boldsymbol{\Sigma}))$, where $\text{tr}(\mathbf{A} \boldsymbol{\Sigma}) = \text{tr}(\mathbf{W}^- \mathbf{W}) = \text{rank}(\mathbf{W})$, because $\mathbf{W}^- \mathbf{W}$ is idempotent with the same rank as $\mathbf{W}$ (Proposition 1.3.3(a)), and the trace of an idempotent matrix is its rank (Proposition 1.9.1).

It remains to show $\text{rank}(\mathbf{W}) = \text{rank}(\mathbf{\Lambda})$. With $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{P}$, $\mathbf{W} = \mathbf{P}^\top \mathbf{M} \mathbf{P} = (\mathbf{M} \mathbf{P})^\top (\mathbf{M} \mathbf{P})$, so $\text{rank} \mathbf{W} = \text{rank}(\mathbf{M} \mathbf{P})$. Also $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{P} = \mathbf{X}^\top \mathbf{M} \mathbf{P}$. If $\mathbf{X}^\top \mathbf{M} \mathbf{P} \mathbf{u} = \mathbf{0}$, then $\mathbf{M} \mathbf{P} \mathbf{u} \in \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X})^\perp = \{\mathbf{0}\}$, so $\mathbf{X}^\top$ is one-to-one on $\mathcal{C}(\mathbf{M} \mathbf{P})$ and $\text{rank} \mathbf{\Lambda} = \text{rank}(\mathbf{M} \mathbf{P})$.

For invariance: $\mathbf{D} \in \mathcal{C}(\mathbf{W})$ with probability one by Theorem 2.2.3(d). If $\mathbf{D} = \mathbf{W} \mathbf{u}$, then $\mathbf{D}^\top \mathbf{W}^- \mathbf{D} = \mathbf{u}^\top \mathbf{W} \mathbf{W}^- \mathbf{W} \mathbf{u} = \mathbf{u}^\top \mathbf{W} \mathbf{u}$, whatever $\mathbf{W}^-$ is. Independence follows from [Theorem 7.5.1\(c\)](#), since $\mathbf{D}$ is a function of $\mathbf{M} \mathbf{Y}$. The F ratio is then a ratio of independent chi-squared variables divided by their degrees of freedom (Definition 4.1.2). ■

The set of $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ values for which this F ratio stays below an F quantile is a confidence ellipsoid. [Chapter 12](#) studies these regions, and [Chapter 11](#) turns the same statistic into the test of the general linear hypothesis. Here we note only that the degrees of freedom in the numerator are the number of *linearly independent* functions being estimated, not the number of rows of $\mathbf{\Lambda}^\top$.

Example. A rank-deficient model, simulated

Consider a one-way layout with three groups of sizes 4, 5, 6, an intercept, one indicator column per group, and a covariate. So $n = 15$ and $p = 5$, but the rank is $r = 4$, because the indicators add up to the intercept. The contrast $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \alpha_1 - \alpha_2$, with $\boldsymbol{\lambda} = (0, 1, -1, 0, 0)^\top$, is estimable. With $\sigma = 2$, [Theorem 7.5.1](#) gives $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} = 1.9007$, computed with the Moore–Penrose inverse. In 100,000 simulated data sets the variance of the estimated contrast is 1.9017, and the average of SSE / σ^2 is 11.001, against $n - r = 11$. The sample correlation between the contrast estimate and s^2 is 0.0036. The standardized contrast T falls within ± 2.201 , the 97.5% point of $t(11)$, in a proportion 0.9498 of the samples. [Figure 7.5.1](#) compares the simulated distributions with the theory. Near the centre the $t(11)$ and $N(0, 1)$ densities are hard to tell apart, so panel (b) uses a logarithmic density scale, on which the tails separate: the simulated points follow $t(11)$, and the normal density falls away

too fast. The tail mass says the same in numbers. A proportion 0.0758 of the samples have $|T| > 1.96$, against 0.05 under $N(0, 1)$ and 0.0758 under $t(11)$. A nominal 95% interval built with the normal point 1.96 would therefore undercover.

```
import numpy as np
from scipy import stats

groups = np.repeat([0, 1, 2], [4, 5, 6])
z = np.array([2.1, 3.4, 1.8, 4.0, 2.9, 3.3, 5.1, 2.2, 3.8, 1.5, 4.4, 2.7, 3.9, 5.0,
              3.1])
n = len(z)
X = np.column_stack([np.ones(n), np.eye(3)[groups], z])  # p = 5 columns, rank 4
r = np.linalg.matrix_rank(X)
G = np.linalg.pinv(X.T @ X)  # one generalized inverse
M = X @ G @ X.T
lam = np.array([0.0, 1.0, -1.0, 0.0, 0.0])  # group 1 minus group 2:
    estimable
beta, sigma = np.array([1.0, 0.5, -0.5, 0.0, 0.8]), 2.0
var_lam = sigma ** 2 * lam @ G @ lam  # Var(lambda^T beta_hat)
print(f"n = {n}, p = {X.shape[1]}, rank = {r}, Var(lambda^T beta_hat) = {var_lam:.4f}
      ")
```

[Run this code online](#)

```
rng = np.random.default_rng(75)
reps = 100_000
Y = X @ beta + sigma * rng.normal(size=(reps, n))  # one data set per row
est = Y @ (X @ G @ lam)  # lambda^T beta_hat =
    lambda^T G X^T y
sse = np.sum((Y - Y @ M) ** 2, axis=1)  # y^T (I - M) y
s2 = sse / (n - r)
t = (est - lam @ beta) / np.sqrt(s2 * (lam @ G @ lam))
print(f"mean {est.mean():.4f}, variance {est.var():.4f}")
print(f"E(SSE)/sigma^2 = {np.mean(sse) / sigma**2:.3f} (n - r = {n - r})")
print(f"corr(estimate, s^2) = {np.corrcoef(est, s2)[0, 1]:.4f}")
print("KS p-value, SSE/sigma^2 vs chi^2(n-r):", round(stats.kstest(sse / sigma**2, "
    chi2", args=(n - r,)).pvalue, 3))
print("KS p-value, t vs t(n-r):", round(stats.kstest(t, "t", args=(n -
    r,)).pvalue, 3))
```

[Run this code online](#)

7.5.4 Without normality

Which parts of [Theorem 7.5.1](#) survive when the errors are merely uncorrelated with common variance? The first two moments of $\lambda^\top \hat{\beta}$ do, by [Theorem 7.2.1](#), and so does $E(s^2) = \sigma^2$. The normality of $\lambda^\top \hat{\beta}$ is lost, but it often returns approximately. $\lambda^\top \hat{\beta}$ is a weighted sum of independent errors, and a central limit theorem applies when no single observation dominates the weights. [Chapter 19](#) makes this precise ([Theorem 19.3.2](#)). The distribution of SSE is less forgiving, because its variance depends on the fourth moment of the errors.

Proposition 7.5.4. Variance of SSE without normality

Let the errors be independent with mean zero, variance σ^2 , third moment μ_3 and fourth

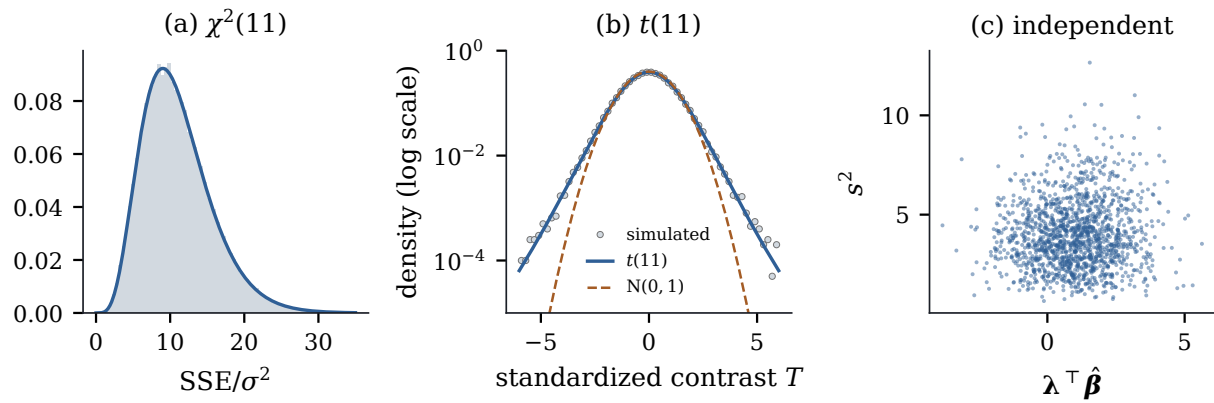


Figure 7.5.1: Sampling distributions in a rank-deficient normal linear model ($n = 15$, $r = 4$; 100,000 simulated data sets). (a) SSE/σ^2 with the $\chi^2(11)$ density. (b) The standardized contrast (binned density, points) with the $t(11)$ and standard normal densities, on a logarithmic scale. (c) The contrast estimate against s^2 : no relationship, as independence requires.

moment μ_4 , and let h_{ii} be the leverages. Then

$$\text{Var}(\text{SSE}) = 2\sigma^4(n - r) + (\mu_4 - 3\sigma^4) \sum_{i=1}^n (1 - h_{ii})^2.$$

Proof. This is Proposition 5.6.3 with r in place of p . That proof applies Theorem 2.3.3 to $\text{SSE} = \mathbf{Y}^\top(\mathbf{I} - \mathbf{M})\mathbf{Y}$ and uses only that $\mathbf{A} = \mathbf{I} - \mathbf{M}$ is idempotent with $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{A}) = n - r$ and diagonal entries $1 - h_{ii}$, and that $\mathbf{A}\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, which kills the terms carrying $\boldsymbol{\theta}$ and μ_3 . None of it needs full column rank. ■

The factor $\mu_4 - 3\sigma^4$ is σ^4 times the excess kurtosis. For normal errors it vanishes and we recover $2\sigma^4(n - r)$. For heavy-tailed errors it is positive, and SSE is more variable than the chi-squared theory says. Intervals for σ^2 built from Theorem 7.5.1(b) are then too short, and this does not improve with sample size: the ratio of the true variance to the nominal one tends to $1 + \frac{1}{2}(\mu_4/\sigma^4 - 3)$ when the leverages are small. The t interval for $\lambda^\top \boldsymbol{\beta}$ is much less affected, because s^2 enters only through s , and s converges to σ whatever the kurtosis.

Example. Laplace errors

Rerun the simulation of Example (A rank-deficient model, simulated) with Laplace (double-exponential) errors of the same variance $\sigma^2 = 4$. For these $\mu_4 = 6\sigma^4$. The contrast still has variance 1.9007. For SSE, $\sum_i (1 - h_{ii})^2 = 8.138$, and Proposition 7.5.4 gives $\text{Var}(\text{SSE}) = 742.6$, more than twice the normal-theory value 352.0. The simulated variance is 741.5. Independence fails too. The contrast and SSE are still uncorrelated, because the errors are symmetric, but the *squared* error of the contrast has correlation 0.211 with SSE. Samples with a few large errors inflate both.

```
E_lap = rng.laplace(scale=sigma / np.sqrt(2), size=(reps, n)) # variance sigma^2,
mu_4 = 6 * sigma^4
Y_lap = X @ beta + E_lap
```

```

est_lap = Y_lap @ (X @ G @ lam)
sse_lap = np.sum((Y_lap - Y_lap @ M) ** 2, axis=1)
h = np.diag(np.eye(n) - M)
var_sse_lap = (6 - 3) * sigma**4 * h @ h + 2 * sigma**4 * (n - r)
print(f"Var(SSE): normal theory {2 * sigma**4 * (n - r):.1f}, Laplace formula {
    var_sse_lap:.1f}, "
      f"simulated {sse_lap.var():.1f}")
print(f"corr(squared error of estimate, SSE) under Laplace errors: "
      f"{np.corrcoef((est_lap - lam @ beta) ** 2, sse_lap)[0, 1]:.3f}")

```

[Run this code online](#)

In the listing, $\mathbf{h}$ holds the diagonal of $\mathbf{I} - \mathbf{M}$, that is, $1 - h_{ii}$.

7.5.5 Exercises

7.5.5.1 A. Check your understanding

Exercise. A1

Under Eq. 7.3.1, find the joint distribution of $(\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}})$. Show that $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$ are independent, and that $\hat{\boldsymbol{\varepsilon}} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$ whatever $\boldsymbol{\beta}$ is. Is $\hat{\boldsymbol{\varepsilon}}$ independent of SSE?

Exercise. A2

In simple linear regression, write out T of [Corollary 7.5.2](#) for the slope, in terms of S_{xx} and s , and state its distribution. What is r ?

7.5.5.2 B. Practice

Exercise. B1

Let $\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2 \in \mathcal{C}(\mathbf{X}^\top)$. Show that $\text{Cov}(\boldsymbol{\lambda}_1^\top \hat{\boldsymbol{\beta}}, \boldsymbol{\lambda}_2^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \boldsymbol{\lambda}_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}_2$, and that under normality the two estimates are independent iff $\boldsymbol{\lambda}_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}_2 = 0$. In a one-way layout with a groups of m observations each and cell-means parameterization $E(Y_{ij}) = \mu_i$, show that the estimates of $\sum_i c_i \mu_i$ and $\sum_i d_i \mu_i$ are independent iff $\sum_i c_i d_i = 0$.

Solution. The covariance is the off-diagonal entry of [Theorem 7.5.1\(a\)](#) with $\boldsymbol{\Lambda} = [\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2]$. Under normality the pair is bivariate normal, so zero covariance is equivalent to independence (Theorem 3.4.1). In the cell-means model $\mathbf{X}^\top \mathbf{X} = m\mathbf{I}_a$, so $\boldsymbol{\lambda}_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}_2 = \sum_i c_i d_i / m$, which vanishes iff $\sum_i c_i d_i = 0$.

Exercise. B2

A new observation $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$ will be taken, with $\varepsilon_0 \sim N(0, \sigma^2)$ independent of $\mathbf{Y}$ and $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$. Show that

$$\frac{Y_0 - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}}{s \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}} \sim t(n - r),$$

and deduce a prediction interval for Y_0 . Why is the extra 1 in the denominator never negligible?

Solution. Y_0 is independent of $\mathbf{Y}$, so $Y_0 - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ is normal with mean $\mathbf{x}_0^\top \boldsymbol{\beta} - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}} = 0$ and variance $\sigma^2 + \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$. It is a function of $(Y_0, \mathbf{M}\mathbf{Y})$, which is independent of SSE. Dividing by $\sigma \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}$ gives a standard normal, independent of $(n - r)s^2/\sigma^2 \sim \chi^2(n - r)$, and the ratio is $t(n - r)$ by Definition 4.1.3. The interval is $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \pm t_{n-r, \alpha/2} s \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}$. The 1 is the variance of the new error itself, which no amount of data about $\boldsymbol{\beta}$ can reduce. The second term tends to zero as the design grows. The first does not.

Exercise. B3

Use Theorem 7.5.1(b) to construct an exact $1 - \alpha$ confidence interval for σ^2 . Then, using Proposition 7.5.4 with small leverages, explain why its coverage does not tend to $1 - \alpha$ as $n \rightarrow \infty$ when the errors have nonzero excess kurtosis.

7.5.5.3 C. Going deeper

Exercise. C1

Give a second proof of Theorem 7.5.1(b) and (c) using Theorem 4.4.4 with the two projections $\mathbf{M}$ and $\mathbf{I} - \mathbf{M}$. Extend it: if $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, show that $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2$ and SSE are independent, and find the distribution of $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2/\sigma^2$ when $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ is arbitrary.

Exercise. C2

For the test that rejects $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = d$ when $|T_d| > t_{n-r, \alpha/2}$, show that the power depends on the parameters only through $|\delta|$ of Corollary 7.5.2. For a fixed design, how does the power change when the sample is replicated m times ($\mathbf{X}$ replaced by m stacked copies)?